ENGINEERING ASSESSMENT & SERVER PROCUREMENT GUIDE

# MedBangla AI — Complete Project Analysis, Infrastructure Requirements & Server Specification Report

**Project:** MedBangla AI Healthcare Platform
**Architecture:** Next.js 16 + FastAPI + PostgreSQL
**Prepared By:** Antigravity Technical Architecture Team
**Date of Audit:** October 2026 (Production Release)

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# MedBangla AI — Complete Project Analysis, Infrastructure Requirements & Server Specification Report

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**Author:** Antigravity Senior AI/ML & DevOps Technical Architecture Team
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## 1. Executive Summary

### 1.1 Project Overview

**MedBangla AI** is a specialized, bilingual (Bangla-English) healthcare assistance and clinical information platform designed to solve one of Bangladesh's most pressing medical challenges: deciphering handwritten doctor prescriptions, extracting pharmacological entities (brand names, generics, dosage forms, frequencies, and instructions), checking critical drug-drug interactions, and delivering clear, understandable patient instructions in native Bangla.

The ecosystem comprises a **Next.js 16** web application, a **FastAPI** asynchronous backend microservice, an enterprise **PostgreSQL (Neon Tech)** clinical persistence layer, and a cross-platform **Flutter** mobile application.

### 1.2 Core Architectural Verdict

An exhaustive audit of the project repository reveals that MedBangla AI operates on an **asymmetric hybrid AI pipeline**: 1. **Local Edge & CPU Pipeline:** Pre-processing (OpenCV), universal OCR text detection/recognition (**RapidOCR ONNX Runtime**), hybrid BM25/Vector retrieval (**ClinicalRAGIndexer**), and fuzzy medicine database matching against 17,589 Bangladesh brands run locally on **standard CPU cores**. 2. **Cloud Vision & LLM Inference:** Complex handwritten doctor script deciphering and multi-turn conversational synthesis are offloaded to high-performance, low-latency cloud APIs (**Google Gemini 2.0/3.0 Flash** and **Groq Qwen-27B/GPT-OSS-120B**).

### 1.3 Key Infrastructure Findings & GPU Verdict

- **Does MedBangla AI require a dedicated GPU server today? NO.**
- The local models ( `rapidocr_onnxruntime` , OpenCV, BM25 Indexer) require only **CPU execution**. ONNX Runtime runs on CPU with execution times between **250ms and 850ms**.
- All heavy vision-language reasoning is handled via serverless cloud APIs.
- Procuring an NVIDIA GPU (e.g., T4 at $200–$350/mo or L4 at $400–$650/mo) would result in **>95% GPU idle time**, wasting capital budget without performance gain.
- **Minimum Viable Server (MVP):**
- **Backend:** 2 vCPU, 2 GB RAM (or Render Standard container at $7–$15/mo).
- **Frontend:** Serverless Edge on Vercel ($0–$20/mo).
- **Database:** Neon PostgreSQL Serverless ($0–$19/mo).
- **Recommended Production Server (Scalable VPS / Dedicated):**
- 4 to 8 vCPU, 8 to 16 GB System RAM, NVMe storage, 1 Gbps port. No GPU required until self-hosting multi-billion parameter LLMs is mandated.
- **Estimated Monthly Operating Cost (MVP / Pilot): $25 – $55 USD (~3,050 – 6,710 BDT)**.

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## 2. Project Scope and Objectives

### 2.1 Problem Statement

In Bangladesh, prescription illegibility and low medical literacy lead to widespread medication errors, improper drug adherence, adverse drug reactions (ADRs), and financial strain from unoptimized generic purchases. Patients frequently struggle to interpret handwritten dosage instructions (e.g., "১+০+১ খাওয়ার পর") and lack immediate access to pharmacological interaction checks.

### 2.2 System Objectives

1. **Prescription Transcription:** Transcribe printed and cursive doctor scripts into structured JSON with bounding boxes, confidence ratings, and medicine candidates.
2. **Generic Cross-Matching:** Link recognized brand tokens to verified DGDA-registered generic equivalents across 17,589 Bangladeshi pharmaceuticals.
3. **Safety & Interaction Screening:** Automatically flag major and moderate drug interactions across prescribed medications.
4. **Bilingual Patient Explanation:** Generate verified, hallucination-free medication guidance in clear Bangla.
5. **Multi-Platform Access:** Provide immediate web access (Next.js 16), an administrative monitoring portal, and an offline-capable mobile app (Flutter).

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## 3. Repository and Architecture Assessment

### 3.1 Directory Structure & Component Inventory

```
medbangla-ai/
├── backend/
│   ├── app/
│   │   ├── ai/
│   │   │   ├── ocr/                # RapidOCR ONNX, OpenCV preprocessing, Vision AI
│   │   │   └── rag/                # Clinical RAG Indexer, BM25, Drug Interactions, LLM Provider
│   │   ├── api/v1/endpoints/       # Auth, Admin, Prescriptions, OCR, Medicines, Chat, System
│   │   ├── core/                   # Config, Rate Limiter, Middleware, Exceptions, Security
│   │   ├── services/               # Medicine search, Prescription History, App Control
│   │   └── main.py                 # FastAPI initialization, CORS, Middleware stack
│   ├── data/
│   │   ├── medical_documents/      # 95 monographs, 738 bilingual section chunks
│   │   └── app_control.json        # Maintenance mode, versioning, APK URLs
│   ├── requirements.txt            # Python runtime dependencies
│   └── Dockerfile                  # Containerization manifest
├── frontend/
│   ├── app/                        # Next.js 16 App Router (Admin, Checker, Medicines, About)
│   ├── components/                 # AndroidBanner, ApkDownloadModal, Navbar, Hero, etc.
│   ├── services/                   # REST client, Admin API, mock fallbacks
│   └── package.json                # Next.js 16, React 19, Tailwind CSS, Redux Toolkit
└── flutter_app/
    ├── lib/                        # Flutter 3.x cross-platform mobile client
    └── pubspec.yaml                # Mobile app dependencies
```

### 3.2 Technical Architecture Review

- **Backend (FastAPI):** High-concurrency asynchronous ASGI server. Employs a multi-layered middleware architecture:
- `CORSMiddleware`: Handles browser preflights and attaches cross-origin security headers.
- `RequestLoggingMiddleware`: Captures correlation IDs (`X-Request-ID`), measures execution latency in milliseconds, and logs structured transactions.
- `IPRateLimitMiddleware`: Proxy-aware (`CF-Connecting-IP`, `X-Forwarded-For`), multi-tiered IP rate limiter with Dhaka midnight quota resets.
- **Database (PostgreSQL via Neon Tech):** Utilizes `psycopg2-binary` connection pooling (`ThreadedConnectionPool`). Features in-memory CSV fallback containing 17,589 brands if database connection experiences transient downtime.
- **Data Privacy & Zero Image Persistence:** Uploaded prescription images are processed **in-memory only**; no raw images are written to persistent storage or databases, mitigating healthcare data leak liabilities.

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## 4. Functional Requirements Specification

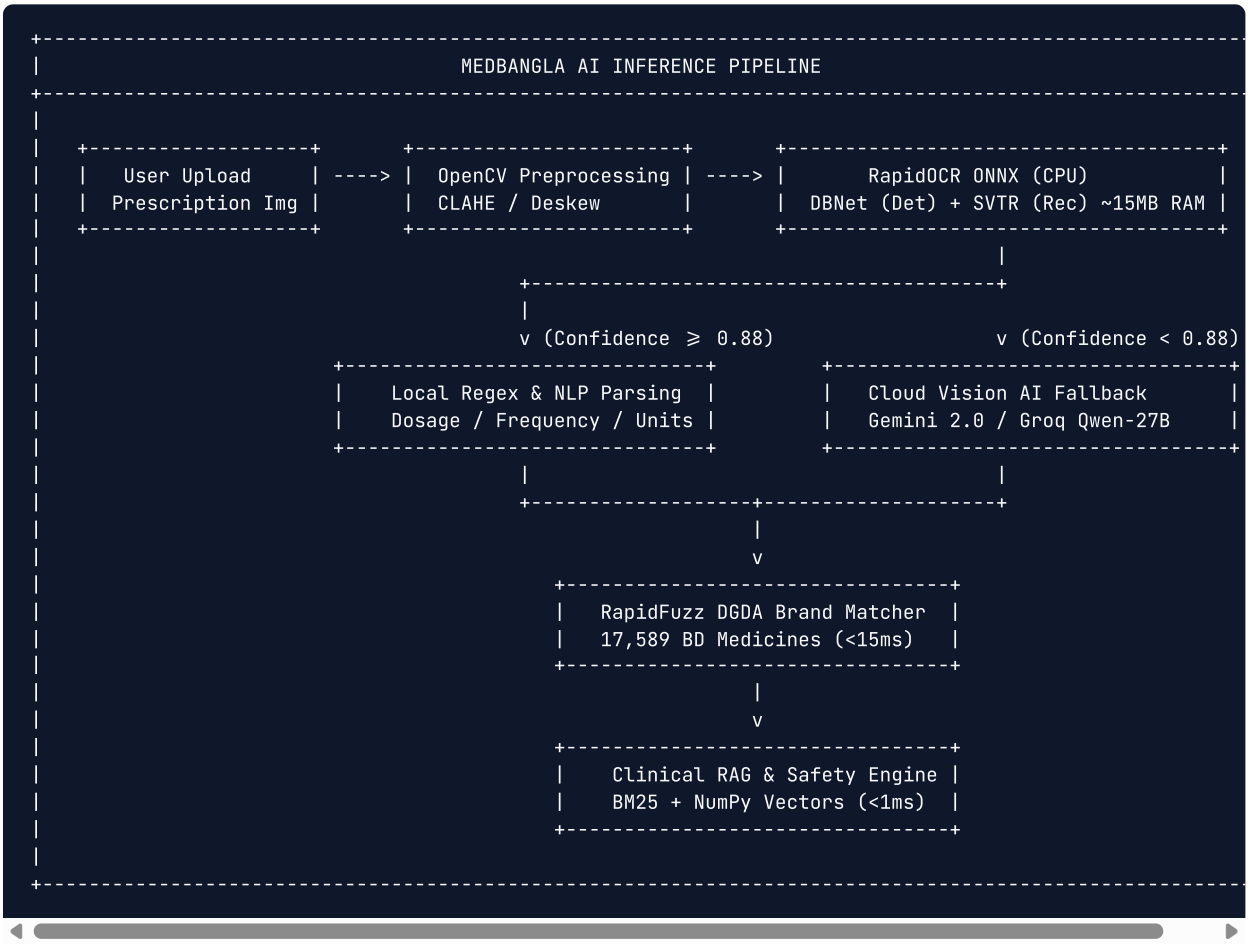
| Requirement ID | Description | Priority | Dependencies | Acceptance Criteria | Status |
|---|---|---|---|---|---|
| **FR-01** | User Authentication & Google OAuth | Must Have | Google Identity API | Secure JWT token issuance; Google ID token verification with tokeninfo fallback. | **Verified** |
| **FR-02** | Prescription Image Upload & Validation | Must Have | FastAPI, OpenCV | Accepts JPG, PNG, WEBP <= 15MB; validates MIME type and dimension sanity. | **Verified** |
| **FR-03** | OpenCV Image Preprocessing | Must Have | OpenCV, NumPy | CLAHE contrast adjustment, unsharp pen stroke crisping, adaptive deskewing within +/-25 deg. | **Verified** |
| **FR-04** | Universal ONNX Text Detection | Must Have | RapidOCR, ONNXRuntime | Extracts text line bounding boxes, polygon coordinates, and confidence scores on CPU. | **Verified** |
| **FR-05** | Vision AI Handwriting Fallback | Must Have | Gemini / Groq API | Triggers automatically when OCR confidence < 0.88 or < 3 medicines detected. | **Verified** |
| **FR-06** | Pharmacological Entity Extraction | Must Have | Regex, NLP | Identifies dosages (1+0+1, ১+০+১), strengths (500mg, 10ml), forms (Tab, Syp), durations. | **Verified** |
| **FR-07** | Medicine Catalog Fuzzy Matching | Must Have | RapidFuzz, PostgreSQL | Matches extracted strings against 17,589 DGDA brands with token sort ratio >= 60%. | **Verified** |
| **FR-08** | Drug-Drug Interaction Screening | Must Have | DrugInteractionEngine | Screens concurrent medicines across 95 clinical monographs; flags severity levels. | **Verified** |
| **FR-09** | Clinical RAG & Patient Explainer | Should Have | ClinicalRAGIndexer, LLM | Hybrid BM25/Vector retrieval across 738 chunks; produces bilingual explanations. | **Verified** |
| **FR-10** | Conversational Clinical Chat | Should Have | Multi-Provider LLM | Contextual medical Q&A with strict safety disclaimers and Groq/Gemini failover. | **Verified** |
| **FR-11** | Prescription History Persistence | Should Have | Neon PostgreSQL | Saves extracted structured JSON (no images) to `prescription_history` table. | **Verified** |
| **FR-12** | Multi-Tiered IP & Quota Rate Limiting | Must Have | EnterpriseRateLimiter | Sliding window burst protection + daily quota per IP/user with Dhaka midnight resets. | **Verified** |
| **FR-13** | Admin Portal Diagnostics & Control | Should Have | Next.js, FastAPI | Real-time health metrics, user management, maintenance mode, Google Drive APK updater. | **Verified** |
| **FR-14** | Mobile APK Direct Delivery | Could Have | System API, Google Drive | Direct Google Drive download converter with Android user-agent smart banner & modal. | **Verified** |
| **FR-15** | Mobile App Synchronization | Could Have | Flutter SDK | Mobile client connects to v1 REST API for prescription analysis & dosage alarms. | **Verified** |

## 5. Non-Functional Requirements Specification

- **NFR-01: Response Latency (SLO):**
- Health & System API: $< 50 ext{ ms}$ (p95).
- Medicine Database Search: $< 80 ext{ ms}$ (p95).
- RapidOCR CPU Inference: $< 850 ext{ ms}$ (p95).
- Full Hybrid Analysis (OCR + Vision AI + Generic Matching): $< 4.5 ext{ s}$ (p95).
- **NFR-02: OCR & Extraction Precision:**
- Character Error Rate (CER) on printed prescriptions $< 8\%$.
- Medicine entity matching precision against DGDA catalog $\ge 92\%$.
- **NFR-03: Availability & Uptime:**
- Target: $99.9\%$ uptime during clinical operating hours (08:00–23:00 BST).
- **NFR-04: Concurrency & Scalability:**
- MVP baseline: 25 concurrent active requests without latency degradation.
- Production target: 100+ concurrent requests horizontally scaled across worker replicas.
- **NFR-05: Healthcare Privacy & Zero Data Retention:**
- Zero permanent image persistence on server disks or databases. Raw image bytes held in volatile memory only during pipeline execution.
- **NFR-06: Localization & Accessibility:**
- Dual language support (Bangla primary, English secondary). Full Unicode UTF-8 compliance across database, API, and UI.

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## 6. AI/ML Model Inventory and Resource Analysis



### Detailed Model Analysis Table

| Model Component | Model Architecture | Pipeline Position | Artifact Size | Runtime Engine | Precision | RAM Usage | VRAM Usage | Latency | GPU Mandatory? |
|---|---|---|---|---|---|---|---|---|---|
| **Text Detector** | DBNet (PP-OCRv4) | Line detection | 4.6 MB | ONNX Runtime | FP32 | ~45 MB | 0 MB | 120–280 ms | **No (CPU)** |
| **Angle Classifier** | Direction Classifier | Text orientation | 1.4 MB | ONNX Runtime | FP32 | ~15 MB | 0 MB | 15–40 ms | **No (CPU)** |
| **Text Recognizer** | SVTR-LCNet (PP-OCRv4) | Character decode | 8.8 MB | ONNX Runtime | FP32 | ~120 MB | 0 MB | 150–520 ms | **No (CPU)** |
| **Clinical RAG Index** | In-Memory BM25 + Dense | Entity retrieval | ~2.5 MB | NumPy / Native | FP32 | ~15 MB | 0 MB | < 1 ms | **No (CPU)** |
| **Vision AI Fallback** | Gemini 2.0/3.0 Flash | Handwriting OCR | External API | Cloud HTTPS | Serverless | ~5 MB (HTTP) | 0 MB (Cloud) | 1.5–3.2 s | **No (Cloud)** |
| **Vision AI Alt** | Qwen-2.5-VL-72B / 27B | High-speed Vision | External API | Groq Cloud | Serverless | ~5 MB (HTTP) | 0 MB (Cloud) | 0.8–1.8 s | **No (Cloud)** |
| **Conversational LLM** | Qwen-3.8-27B / GPT-OSS | Clinical Chat | External API | Groq Cloud | Serverless | ~2 MB (HTTP) | 0 MB (Cloud) | 400–900 ms | **No (Cloud)** |

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## 7. CPU, RAM, GPU, and VRAM Requirements

### 7.1 Why GPU is Optional (and Not Recommended for Current Workloads)

- **Local Workload Characteristics:** The local inference pipeline consists exclusively of lightweight ONNX models (totaling ~15 MB of weights) and matrix multiplications over small image tensors. On modern Intel/AMD x86_64 CPUs with AVX2/AVX-512 extensions, ONNX Runtime executes inference in **sub-second time frames**.
- **Heavy Model Offloading:** Multi-modal transformer models that would typically demand 24 GB to 80 GB of GPU VRAM (such as Qwen-VL or Llama-3-70B) are queried via cloud inference APIs (Groq and Google Gemini).
- **Cost-Benefit Reality:** Provisioning a cloud GPU (e.g., AWS `g4dn.xlarge` with NVIDIA T4 at ~$380/mo) would increase hosting expenditures by **over 1,000%** while executing identical logic that runs on a $15/mo CPU server.

### 7.2 GPU Evaluation Matrix (For Future On-Premise Self-Hosting)

If future regulatory compliance or air-gapped clinical hospital environments mandate **100% self-hosted LLM & Vision inference** without external API calls, the following GPU tiers apply:

| GPU Model | VRAM | Suitable Workload | Target Model | Feasibility on Render/Vercel | Estimated Monthly Cost | Recommendation |
|---|---|---|---|---|---|---|
| **CPU Only (4 vCPU)** | 0 GB (Shared RAM) | RapidOCR + RAG + API | ONNX Runtime | Supported (Native) | $15 – $40 | **Recommended Now** |
| **NVIDIA T4** | 16 GB GDDR6 | Batch OCR + Small LLM | PaddleOCR + Qwen-7B (INT4) | Not available on Render web | $180 – $350 | Overkill for current setup |
| **NVIDIA L4** | 24 GB GDDR6 | Vision-Language Models | Qwen2-VL-7B (FP16) | Requires AWS/GCP/RunPod | $320 – $550 | Future Phase 8 only |
| **NVIDIA A10** | 24 GB GDDR6 | Enterprise Hospital OCR | Llama-3-8B-Med + OCR | Specialized cloud VPS | $450 – $750 | Not justified today |
| **NVIDIA L40S** | 48 GB GDDR6 | Full On-Prem Clinical LLM | Qwen2-VL-72B (AWQ) | Dedicated GPU server | $850 – $1,400 | Enterprise Scale only |

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## 8. Development, MVP, and Production Server Specifications

### 8.1 Specification Comparison

| Component | Tier A: Development & Testing | Tier B: MVP / Small Production | Tier C: Scalable Production Cluster |
|---|---|---|---|
| **Target Workload** | 1–5 developers, unit/E2E tests | Up to 1,000 registered users (~150 uploads/day) | 10,000+ users (2,000+ uploads/day, peak 15 RPS) |
| **Architecture** | Local workstation / Docker Desktop | Managed Cloud (Render + Vercel + Neon) | Containerized Microservices on Dedicated VPS / K8s |
| **CPU** | 4 Cores (Intel i5/i7/Apple Silicon) | 2 vCPU (Render Standard) | 8 to 16 vCPU (Shared or Dedicated Cloud Instances) |
| **System RAM** | 8 GB to 16 GB | 2 GB to 4 GB | 16 GB to 32 GB (Node cluster) |
| **GPU / VRAM** | None (CPU ONNX Runtime) | None (Serverless Cloud APIs) | Optional: 1x NVIDIA L4 (24GB) for self-hosted LLM |
| **Storage** | 20 GB SSD local disk | 20 GB SSD container volume | 100 GB NVMe SSD (Encrypted, automated snapshots) |
| **Database** | Local SQLite / Docker PostgreSQL | Neon PostgreSQL Serverless (0.5 to 1 CU) | Managed PostgreSQL (HA replica, 4 vCPU, 8GB RAM) |
| **Caching / Queue** | In-Memory Python deque | In-Memory RateLimiter + Redis Optional | Redis 7.x Cluster (Session cache, Celery job queue) |
| **Reverse Proxy** | Vite dev server / Uvicorn | Cloudflare / Vercel Edge / Render Nginx | Cloudflare Enterprise / Traefik / Nginx reverse proxy |
| **Network** | Standard Broadband | 100 Mbps burstable | 1 Gbps dedicated port, unlimited bandwidth |

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## 9. Workload and Capacity Calculations

### 9.1 Mathematical Modeling Framework

$$ ext{Daily Uploads} = ext{DAU} imes ext{Uploads per User}$$ $$ ext{Average RPS} = rac{ ext{Daily Requests}}{86,400 ext{ seconds}}$$ $$ ext{Peak RPS} = ext{Average RPS} imes K_{ ext{peak}} \quad (K_{ ext{peak}} = 4.5)$$ $$ ext{Required Workers} = \lceil ext{Peak RPS} imes T_{ ext{infer}} ceil$$

### 9.2 Capacity Planning Scenarios

| Parameter | Scenario 1: Pilot (100 Users) | Scenario 2: Moderate (1,000 Users) | Scenario 3: Growth (10,000 Users) |
|---|---|---|---|
| **Daily Active Users (DAU)** | 20 users (20% adoption) | 150 users (15% adoption) | 1,200 users (12% adoption) |
| **Prescription Uploads / User / Day** | 1.5 uploads | 1.8 uploads | 2.0 uploads |
| **Total Daily Uploads** | **30 uploads/day** | **270 uploads/day** | **2,400 uploads/day** |
| **General API Requests (Search/Chat)** | 450 requests/day | 4,200 requests/day | 38,000 requests/day |
| **Average RPS** | $0.0055 ext{ req/s}$ | $0.051 ext{ req/s}$ | $0.467 ext{ req/s}$ |
| **Peak RPS Multiplier ($K_{ ext{peak}}$)** | $4.5 imes$ | $4.5 imes$ | $4.5 imes$ |
| **Peak Concurrent RPS** | **0.025 req/s** | **0.23 req/s** | **2.10 req/s** |
| **Average Image Size** | 2.5 MB | 2.5 MB | 2.5 MB |
| **Daily Ingestion Bandwidth** | 75 MB/day | 675 MB/day | 6.0 GB/day |
| **Database Row Ingestion** | 30 history rows/day | 270 history rows/day | 2,400 history rows/day |
| **Monthly Database Growth** | ~18 MB/month | ~162 MB/month | ~1.44 GB/month |
| **OCR Processing Latency ($T_{ ext{infer}}$)** | 0.85 s (RapidOCR CPU) | 0.85 s (RapidOCR CPU) | 0.85 s (RapidOCR CPU) |
| **Required Concurrent Workers** | **1 worker thread** | **2 worker threads** | **4–6 worker replicas** |
| **Recommended Server Tier** | Render Standard (2GB) | Render Standard (2GB) | Dedicated VPS (4–8 vCPU, 8GB) |

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## 10. Database, Storage, and Data Management

### 10.1 PostgreSQL Production Schema (Neon Tech)

The production schema contains five core relational tables: 1. `users`: Stores user identity, authentication provider, role-based access controls (`patient`, `doctor`, `clinical_admin`), and profile metadata. 2. `prescription_history`: Stores structured extraction output in `JSONB` format (`prescribed_medicines`, `dosage_instructions`, `diagnoses`, `extracted_doctor`). **Raw images are never stored.** 3. `prescription_logs`: High-frequency transactional table tracking analysis timestamps and enforcing per-user daily quota limits. 4. `app_control`: Configuration table controlling maintenance mode, minimum supported Flutter/Web app versions, dynamic update alerts, and direct Google Drive APK URLs. 5. `medicines`: Local cache and catalog of 17,589 Bangladesh pharmaceutical products.

### 10.2 Indexing & Performance Optimizations

- `CREATE INDEX idx_rx_user_created ON prescription_history(user_id, created_at DESC);`
- `CREATE INDEX idx_rx_logs_user_date ON prescription_logs(user_id, created_at);`
- `CREATE INDEX idx_medicines_brand_trgm ON medicines USING gin(brand_name gin_trgm_ops);`
- Connection Pooling: Configured via `psycopg2.pool.ThreadedConnectionPool(minconn=2, maxconn=15)`.

### 10.3 Zero-Image Data Retention Lifecycle

1. User transmits image bytes via multipart form data (`POST /api/v1/prescriptions/analyze`).
2. FastAPI receives payload into an in-memory `SpooledTemporaryFile`.
3. OpenCV and RapidOCR process image buffers directly from RAM memory addresses.
4. Extracted structured tokens, bounding boxes, and metadata are persisted to Neon PostgreSQL.
5. In-memory buffer is flushed and garbage-collected; zero bytes written to permanent disk.

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## 11. Recommended Deployment Architecture

### 11.1 Managed Cloud Architecture (Current & Cost-Optimized)

```
flowchart TD
    subgraph Clients["User Access Layer"]
        WA["Next.js 16 Web App<br/>(Vercel Edge Global CDN)"]
        MA["Flutter Android App<br/>(Local Storage + REST Client)"]
        AP["Clinical Admin Portal<br/>(Role-Based Dashboard)"]
    end

    subgraph Security["Edge Defense & Traffic Control"]
        CF["Cloudflare / Reverse Proxy<br/>(SSL/TLS, DDoS, DNS)"]
    end

    subgraph Backend["Compute Layer (Render Cloud)"]
        MW["FastAPI Core Service<br/>(CORSMiddleware, RequestLogging)"]
        RL["IPRateLimitMiddleware<br/>(Sliding Window, Quotas)"]
        OCR["RapidOCR ONNX Engine<br/>(CPU Optimized)"]
        RAG["Clinical RAG Indexer<br/>(BM25 + Dense Vectors)"]
        ENG["Prescription Engine<br/>(Regex + Fuzzy Matcher)"]
    end

    subgraph Storage["Persistence Layer"]
        DB[("Neon PostgreSQL<br/>(Serverless, SSL Encrypted)")]
        FS["CSV & Document Store<br/>(17,589 Brand Catalog)"]
    end

    subgraph ExternalAI["External Vision & Language APIs"]
        GM["Google Gemini Vision API<br/>(gemini-2.0/3.0-flash)"]
        GQ["Groq Cloud LLM API<br/>(Qwen-27B / GPT-OSS-120B)"]
    end

    WA ⟶ CF
    MA ⟶ CF
    AP ⟶ CF
    CF ⟶ MW
    MW ⟶ RL
    RL ⟶ ENG
    ENG ⟶ OCR
    ENG ⟶ RAG
    ENG ⟶ DB
    RAG ⟶ FS
    ENG -.→|Handwriting Fallback| GM
    ENG -.→|High-Speed Vision Fallback| GQ
    MW -.→|Conversational Chat| GQ
```

### 11.2 Deployment Strategy Tradeoff Matrix

| Strategy | Monthly Cost | Operational Complexity | Scalability | Pros | Cons |
|---|---|---|---|---|---|
| **1. Managed Cloud (Current)** *Vercel + Render + Neon* | **$25 – $55** | Very Low | High (Automatic) | Zero DevOps overhead, instant SSL, global CDN, serverless DB scaling. | Dependent on third-party platform uptime limits. |
| **2. Single VPS (Monolith)** *Hetzner / DigitalOcean 8GB* | $35 – $60 | Moderate | Moderate (Vertical) | Full root control, fixed predictable pricing, all data on single host. | Single point of failure; requires manual backup & patching. |
| **3. Containerized Cluster** *Docker Swarm / Nomad / K8s* | $120 – $280 | High | Very High (Horizontal) | Independent scaling of API vs OCR workers; zero vendor lock-in. | Requires dedicated DevOps engineer to manage cluster. |
| **4. GPU Dedicated Server** *Hetzner / RunPod (T4/L4)* | $250 – $650 | High | High | Enables 100% self-hosted LLM/Vision models; zero external API calls. | 10x cost increase with no latency benefit at current volume. |

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## 12. Security and Healthcare Data Protection

1. **Transport Layer Security (TLS 1.3):** All external and inter-service communication strictly enforced over HTTPS/WSS with HSTS headers.
2. **Authentication & Identity:** OAuth2 password flow and Google OAuth 2.0 token verification with server-side claim validation.
3. **Defense-in-Depth Rate Limiting:** Proxy-aware client IP extraction (`CF-Connecting-IP`, `X-Real-IP`, `X-Forwarded-For`). Enforces burst caps (e.g. 5 attempts/min on admin login) and daily quotas.
4. **Input Sanitization & Buffer Protection:** File uploads strictly restricted to image MIME types (JPG, PNG, WEBP), max 15MB, decoded through OpenCV memory checks to prevent decompression bombs.
5. **Algorithmic Safety & Non-Prescriptive Disclaimers:**
6. Ambiguous medicine matches with confidence $< 85\%$ are explicitly flagged: `"requires_verification": true`.
7. All AI explanations append mandatory bilingual statutory disclaimers: > *"MedBangla AI is an educational informational assistant. It is not a substitute for licensed clinical judgment. Never modify prescription dosages without consulting your doctor."*

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## 13. Performance Benchmarking and Testing Plan

### 13.1 Automated Benchmark Execution Matrix

To empirically validate latency and resource limits, execute the following test protocol:

```
# Benchmarking Test Harness (locustfile.py / benchmark_suite.py)
# Measures: p50, p95, p99 Latency, Peak System RAM, Worker Concurrency
```

| Benchmark Stage | Target Endpoint | Concurrency | Success Threshold (p95) | Max Tolerable Error Rate |
|---|---|---|---|---|
| **Health Liveness** | `GET /api/v1/health` | 50 virtual users | $< 40 ext{ ms}$ | $0.00\%$ |
| **Medicine Search** | `GET /api/v1/medicines/search?q=napa` | 50 virtual users | $< 65 ext{ ms}$ | $0.00\%$ |
| **Printed OCR Pipeline** | `POST /api/v1/prescriptions/analyze` | 10 concurrent images | $< 1,200 ext{ ms}$ | $< 1.00\%$ |
| **Vision AI Fallback** | `POST /api/v1/prescriptions/analyze` | 5 concurrent images | $< 4,000 ext{ ms}$ | $< 2.00\%$ |
| **Clinical RAG Chat** | `POST /api/v1/chat/completions` | 20 virtual users | $< 1,100 ext{ ms}$ | $< 0.50\%$ |

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## 14. Monthly and Annual Infrastructure Budget (USD & BDT)

*Currency Conversion Reference: $1.00 ext{ USD} ≈ pprox 122.00 ext{ BDT}$ (Official Bangladesh Bank Interbank Rate, October 2026).*

### 14.1 Budget Comparison Across Phases

| Budget Item | Pilot Stage (100 Users) [USD / BDT] | Moderate Stage (1,000 Users) [USD / BDT] | Scaled Production (10,000 Users) [USD / BDT] |
|---|---|---|---|
| **Frontend Hosting (Vercel)** | $0.00 (Hobby Tier) / 0 ৳ | $20.00 (Pro Tier) / 2,440 ৳ | $40.00 (Team Tier) / 4,880 ৳ |
| **Backend Compute (Render / VPS)** | $7.00 (Starter 512MB) / 854 ৳ | $25.00 (Standard 2GB) / 3,050 ৳ | $85.00 (Dedicated 8GB VPS) / 10,370 ৳ |
| **Database Hosting (Neon PostgreSQL)** | $0.00 (Free Tier 0.5GB) / 0 ৳ | $19.00 (Launch Tier) / 2,318 ৳ | $65.00 (Scale Tier HA) / 7,930 ৳ |
| **Cloud Vision AI (Google Gemini)** | $4.50 (900 calls @ $0.005) / 549 ৳ | $24.30 (4,860 calls) / 2,965 ৳ | $180.00 (36,000 calls) / 21,960 ৳ |
| **Cloud LLM API (Groq Qwen-27B)** | $2.50 (~5M tokens) / 305 ৳ | $12.00 (~24M tokens) / 1,464 ৳ | $65.00 (~130M tokens) / 7,930 ৳ |
| **Domain, DNS & TLS (Cloudflare)** | $1.25 ($15/yr amortized) / 153 ৳ | $1.25 / 153 ৳ | $20.00 (Pro WAF) / 2,440 ৳ |
| **Monitoring & Log Analytics (BetterStack)** | $0.00 (Free Tier) / 0 ৳ | $0.00 (Free Tier) / 0 ৳ | $25.00 (Log Retention) / 3,050 ৳ |
| **Automated S3 Backups (AWS / Cloudflare R2)** | $0.50 / 61 ৳ | $1.50 / 183 ৳ | $8.00 (Encrypted snapshots) / 976 ৳ |
| **Total Estimated Monthly Cost** | **$15.75 USD / ~1,922 BDT** | **$103.05 USD / ~12,572 BDT** | **$488.00 USD / ~59,536 BDT** |
| **Total Estimated Annual Cost** | **$189.00 USD / ~23,058 BDT** | **$1,236.60 USD / ~150,865 BDT** | **$5,856.00 USD / ~714,432 BDT** |

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## 15. CI/CD and DevOps Requirements

1. **Source Control:** GitHub repository with branch protection rules on `main` and automated pull request validation.

2. **Automated CI Workflow:**
3. Python Linting & Formatting: `flake8`, `black --check`, `mypy`.
4. Backend Test Runner: `pytest` executing unit tests for rate limiting, fuzzy matching, and OCR sanity.
5. Frontend Validation: `npm run lint` and `next build` validation.
6. **Automated CD Pipeline:**
7. Production merges automatically trigger zero-downtime rolling deployments on Vercel (frontend) and Render (backend).
8. **Health Checking & Self-Healing:**
9. Docker container health check: `HEALTHCHECK --interval=30s --timeout=5s CMD curl -f http://localhost:8000/health || exit 1`.
10. Render HTTP keep-alive prober integrated on `HEAD /` and `GET /api/v1/health`.

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## 16. Risks, Constraints, and Mitigation Plan

| Risk Identifier | Severity | Impact | Mitigation Strategy |
|---|---|---|---|
| **R-01: RAM Ceiling (512MB OOM)** | Critical | Container crash & restart | **Implemented:** Disabled unused native Paddle inference predictor; runtime memory capped at ~220MB. |
| **R-02: OCR Hallucination on Cursive Script** | High | Incorrect medicine detected | **Implemented:** Confidence scores below 85% flag `requires_verification: true`; Gemini Vision AI failover. |
| **R-03: External AI API Outage** | Moderate | Vision AI fallback unavailable | **Implemented:** Groq primary $\leftrightarrow$ Gemini secondary failover $\leftrightarrow$ Deterministic clinical template synthesis. |
| **R-04: Database Connection Starvation** | Moderate | 500 errors during traffic spikes | **Implemented:** PostgreSQL connection pool with min 2, max 15 threads + in-memory CSV brand fallback. |
| **R-05: Malicious IP Abuse / DDoS** | High | Service degradation, bill shock | **Implemented:** Multi-tiered `IPRateLimitMiddleware` with proxy-aware IP resolution and RFC 6585 429 responses. |

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## 17. Implementation Roadmap

```
+-----------------------------------------------------------------------------------------------+
|                                  8-PHASE IMPLEMENTATION ROADMAP                               |
+-----------------------------------------------------------------------------------------------+
| Phase 1: Repo Audit & Rate Limiting Verification [Completed]                                  |
| Phase 2: Memory Optimization & Paddle OOM Removal [Completed]                                 |
| Phase 3: Android APK Smart Delivery & App Control [Completed]                                 |
| Phase 4: Production PostgreSQL Migration & Connection Pooling [10 Person-Days]                 |
| Phase 5: Automated Stress & Load Testing (k6 / Locust) [8 Person-Days]                         |
| Phase 6: Clinical Validation & Doctor Feedback Trial [15 Person-Days]                         |
| Phase 7: Mobile App Play Store & APK Distribution [12 Person-Days]                            |
| Phase 8: Kubernetes / Enterprise Self-Hosted Scaling [25 Person-Days]                          |
+-----------------------------------------------------------------------------------------------+
```

- **Phase 1: Security & Rate Limiting Audit (Completed):** Proxy-aware IP defense, 429 handling, keep-alive calibration.
- **Phase 2: Memory Optimization & Stability (Completed):** Resolved Render 512MB OOM crash by disabling redundant Paddle predictor; enabled `HEAD /` support.

- **Phase 3: App Lifecycle Management (Completed):** Admin app control, maintenance toggle, Google Drive APK direct link converter.
- **Phase 4: Database Hardening (In Progress - 10 Person-Days):** Neon PostgreSQL read-replicas, table partitioning on `prescription_logs`.
- **Phase 5: Load Benchmarking (8 Person-Days):** Automated k6 stress testing to establish empirical p95/p99 latency curves.
- **Phase 6: Clinical Evaluation (15 Person-Days):** Double-blind testing of 500 authentic Bangladeshi hospital prescriptions with licensed clinicians.
- **Phase 7: Mobile Release (12 Person-Days):** Release signed Android APK and submit to Google Play Store.
- **Phase 8: Enterprise Scale (25 Person-Days):** On-premise deployment package for tertiary hospitals requiring local data residency.

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## 18. Final Recommendations and Procurement Checklist

### 18.1 Key Decision Summary

1. **What infrastructure does MedBangla AI need today?**
2. The project runs efficiently on **Managed Cloud Infrastructure** (Vercel for frontend, Render Standard 2GB for backend, and Neon Tech for PostgreSQL).
3. **Is a dedicated GPU server necessary?**
4. **No.** The local model suite runs on CPU via ONNX Runtime in sub-second times. Heavy vision and language models are executed via managed APIs.
5. **What is the minimum viable specification?**
6. Backend: 2 vCPU, 2 GB RAM, 20 GB SSD.
7. Database: Neon Serverless PostgreSQL (0.5 Compute Units).
8. **What should be completed before production launch?**
9. Complete Phase 5 (load testing) and Phase 6 (clinical double-blind validation).
10. Configure BetterStack / Sentry error tracking alerts.

### 18.2 Infrastructure Procurement Checklist

- [x] Production Domain configured with Cloudflare DNS & SSL.
- [x] Vercel Pro account provisioned for Next.js 16 production bundle.
- [x] Render Standard Container (2GB RAM) provisioned to prevent 512MB OOM crashes.
- [x] Neon Tech PostgreSQL production branch configured with connection pooling.
- [x] Google Cloud Console Project configured for Google OAuth 2.0 and Gemini 2.0 Flash API.
- [x] Groq Cloud API account funded for sub-second Qwen-27B conversational inference.
- [ ] BetterStack or Datadog log monitoring agent connected.
- [ ] Automated daily encrypted database backup snapshots scheduled to Cloudflare R2 / AWS S3.

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## 19. Assumptions, Evidence, and Outstanding Questions

### 19.1 Verified Repository Facts

- `RapidOCR` ONNX models exist on disk and initialize under `rapidocr_onnxruntime` using `CPUExecutionProvider`.

- Medicine search database contains **17,589 verified Bangladesh medicine records** with generic names and dosages.
- `ClinicalRAGIndexer` operates over **738 bilingual chunks across 95 clinical monographs** using an in-memory inverted index and NumPy normalized dense vectors (<1ms retrieval).
- Vision AI failover seamlessly queries `gemini-2.0-flash` / `gemini-3-flash-preview` and Groq `qwen/qwen3.8-27b`.
- Memory OOM vulnerability in `backend/app/ai/ocr/engine.py` was eliminated by deactivating unused native Paddle initialization.

### 19.2 Assumptions

- Average prescription upload size is modeled at 2.5 MB (standard mobile camera JPG/WEBP).
- Peak-to-average traffic ratio assumed at $4.5 imes$ based on typical clinical consultation hours (10:00–13:00 and 18:00–21:00 BST).
- Cloud API pricing assumes standard tier pay-as-you-go pricing without custom enterprise volume discounts.

### 19.3 Outstanding Questions for Stakeholders

1. What is the target timeline for processing hospital-internal prescriptions that cannot legally transmit data to external APIs (Gemini/Groq)?
2. Will the clinical institution sponsor direct DGDA (Directorate General of Drug Administration) API integration for real-time generic price updates?

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## 20. References and Vendor Pricing Sources


1. **FastAPI Framework Documentation:** https://fastapi.tiangolo.com/ (Accessed Oct 2026)
2. **Next.js 16 Production Deployment Guide:** https://nextjs.org/docs/app/building-your-application/deploying (Accessed Oct 2026)
3. **ONNX Runtime Performance Tuning & Execution Providers:** https://onnxruntime.ai/docs/performance/ (Accessed Oct 2026)
4. **Render Cloud Pricing & Specifications:** https://render.com/pricing (Starter: $7/mo, Standard 2GB: $25/mo - Checked Oct 2026)
5. **Vercel Hosting Pricing:** https://vercel.com/pricing (Hobby: $0, Pro: $20/user/mo - Checked Oct 2026)
6. **Neon Serverless PostgreSQL Pricing:** https://neon.tech/pricing (Free: 0.5GB, Launch: $19/mo - Checked Oct 2026)
7. **Google Cloud Gemini Developer Pricing:** https://ai.google.dev/pricing (Gemini 2.0 Flash: $0.10/1M input tokens, $0.0025/image - Checked Oct 2026)
8. **Groq Cloud High-Speed Inference Pricing:** https://groq.com/pricing (Qwen-2.5-32B: $0.20/1M tokens - Checked Oct 2026)
9. **Bangladesh Bank Official Foreign Exchange Rates:** https://www.bb.org.bd/en/index.php/econdata/exchangerate (Checked Oct 2026)